Vectors and the scalar product

Lecture and worked examples

Maths Explorer

Chapter path

Objectives. Choose between projection, coordinates and angle; calculate a length; characterise orthogonality and a locus.

Prerequisites. Vectors, Chasles's relation, Pythagoras, cosine in a right triangle, orthonormal frame.

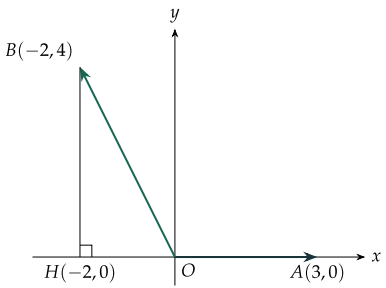
Yearly reference point. 6-15 October: 8 hours

Projection and angle

For two nonzero vectors $\vec{u}, \vec{v}$, their scalar product is the real number $\vec{u} \cdot \vec{v} = \left\|\vec{u}\right\| \left\|\vec{v}\right\| \cos\theta$, where $\theta \in [0, \pi]$ is their unoriented angle. With a zero vector, the product is 0. If H is the projection of B onto (OA), $\overrightarrow{OA} \cdot \overrightarrow{OB}$ equals $OA \times OH$ when H is on the ray $[OA)$, and its opposite otherwise.

Projection and angle

For two vectors of norms 3 and 4 forming 60°, the product is $\frac{3 \times 4 \times 1}{2} = 6$. With 120°, it is -6. If $A(3,0)$ and $B(-2,4)$, the projection is $H(-2,0)$ and the product is -6.



Projection and angle

The result is a number, not a vector. A right angle corresponds to 0, not to 1.

Projection and angle

Two vectors have norms 2 and 5 and form 90°, then 180°. Calculate the products.

Coordinates, norm and orthogonality

In an orthonormal basis, for $\vec{u} = (x, y)$ and $\vec{v} = (x', y')$, $\vec{u} \cdot \vec{v} = xx' + yy'$ and $\left\|\vec{u}\right\|^2 = x^2 + y^2$. The vectors are orthogonal if and only if their product is zero. For an orthonormal basis $(\vec{i}, \vec{j})$, $x = \vec{u} \cdot \vec{i}$ and $y = \vec{u} \cdot \vec{j}$.

Coordinates, norm and orthogonality

$\vec{u} = (2, 3)$, $\vec{v} = (3, -2)$: product $6 - 6 = 0$. The two vectors are orthogonal. Their norm is $\sqrt{13}$. The product of $(2, 3)$ with $(1, 0)$ is 2: we recover the x-coordinate.

Coordinates, norm and orthogonality

The usual formula assumes an orthonormal basis. Collinearity and orthogonality are different properties.

Coordinates, norm and orthogonality

For which real number t are the vectors $(t, 2)$ and $(3, -6)$ orthogonal?

Vector operations and identities

The scalar product is symmetric and bilinear: $\vec{u} \cdot (\vec{v} + \vec{w}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w}$ and $(\lambda\vec{u}) \cdot \vec{v} = \lambda(\vec{u} \cdot \vec{v})$. Hence $\left\|\vec{u} \pm \vec{v}\right\|^2 = \left\|\vec{u}\right\|^2 \pm 2\vec{u} \cdot \vec{v} + \left\|\vec{v}\right\|^2$. It follows that

$$\vec{u} \cdot \vec{v} = \frac{\left\|\vec{u} + \vec{v}\right\|^2 - \left\|\vec{u}\right\|^2 - \left\|\vec{v}\right\|^2}{2}.$$

Vector operations and identities

If $\left\|\vec{u}\right\| = 3$, $\left\|\vec{v}\right\| = 4$ and $\left\|\vec{u}+\vec{v}\right\| = 5$, the product is $\dfrac{25-9-16}{2} = 0$. We deduce orthogonality without coordinates.

Vector operations and identities

In general, $\left\|\vec{u}+\vec{v}\right\| \neq \left\|\vec{u}\right\| + \left\|\vec{v}\right\|$. The identities concern the squares of the norms.

Vector operations and identities

Given $\left\|\vec{u}\right\| = 2$, $\left\|\vec{v}\right\| = 3$, $\vec{u} \cdot \vec{v} = 1$. Calculate $\left\|\vec{u} - \vec{v}\right\|$.

Al-Kashi and choosing a method

In a triangle ABC, $BC^2 = AB^2 + AC^2 - 2AB\,AC\cos\widehat{BAC}$. This formula allows us to calculate the third side from two sides and the included angle, or to find the cosine of an angle from the three sides.

Al-Kashi and choosing a method

$AB = 5$, $AC = 3$ and $\widehat{BAC} = 60°$: $BC^2 = 25 + 9 - 15 = 19$, so $BC = \sqrt{19}$. With three sides, isolate the cosine before approximating the angle on a calculator, if one is allowed.

Al-Kashi and choosing a method

The angle is included between the two sides of the product term. The minus sign is essential.

Al-Kashi and choosing a method

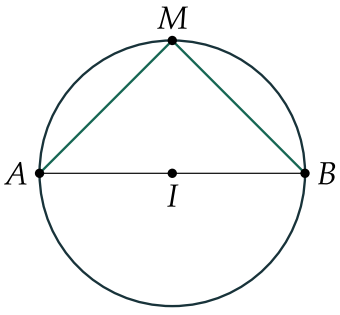
A triangle has $AB = AC = 4$ and $BC = 4\sqrt{2}$. Determine the angle A.

The circle with diameter AB

For two distinct points A and B with midpoint I, $\overrightarrow{MA} \cdot \overrightarrow{MB} = MI^2 - \frac{AB^2}{4}$. The set of points M such that $\overrightarrow{MA} \cdot \overrightarrow{MB} = 0$ is the circle with centre I and radius $\frac{AB}{2}$, including the points A and B.

The circle with diameter AB

$A(-2,0)$ and $B(2,0)$: I is the origin. The locus is $x^2 + y^2 = 4$. For $M(0,1)$, the product is $1 - 4 = -3$; for $M(0,2)$, it is 0.



The circle with diameter AB

The endpoints belong to the locus even though “right triangle” does not apply to a degenerate triangle.

The circle with diameter AB

For $A(0,0)$, $B(6,0)$, determine the points M such that $\overrightarrow{MA} \cdot \overrightarrow{MB} = 7$.